

SYLVA MAGAZINE

■■■■■■■■
Technical Handbook Extended

VOLUME II

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2026■4■

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Volume I

McMillan • Eliashberg • •
Lean • •

8 McMillan

8.1

$$H_{ep} = \sum_{\mathbf{k}, \mathbf{q}, \sigma} g_{\mathbf{k}, \mathbf{q}} (b_{\mathbf{q}} + b_{-\mathbf{q}}^\dagger) c_{\mathbf{k}+\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}, \sigma}$$

$$g_{\mathbf{k}, \mathbf{q}} = \sum_{\sigma} \langle b_{\mathbf{q}} | c_{\mathbf{k}, \sigma} | \rangle$$

8.2

Matsubara

$$\Sigma(\mathbf{k}, i\omega_n) = -T/N \sum_{\mathbf{k}', n'} V_{\text{eff}}(\mathbf{k}-\mathbf{k}', i\omega_n - i\omega_{n'}) G(\mathbf{k}', i\omega_{n'})$$

$$V_{\text{eff}} = G^{-1}$$

8.3

$\Delta(\mathbf{k})$ BCS

$$\Delta(\mathbf{k}) = -\sum_{\mathbf{k}'} V_{\mathbf{k}, \mathbf{k}'} \Delta(\mathbf{k}') / (2E_{\mathbf{k}'}) \tanh(E_{\mathbf{k}'}/2T)$$

$$E_{\mathbf{k}} = \sqrt{\epsilon_{\mathbf{k}}^2 + \Delta_{\mathbf{k}}^2}$$

8.4 McMillan

$$T_c = (\Theta_D/1.45) \exp[-1.04(1+\lambda)/(\lambda - \mu^* - 0.62\lambda\mu^*)]$$

$$\lambda = \frac{2}{\Theta_D} \int_0^{\Theta_D} d\omega \frac{\alpha^2 F(\omega)}{\omega} \quad \mu^* = \frac{\mu}{1 + \mu \ln(\epsilon_F/\omega_c)}$$

8.5

1. ■■■■■■■■■■■■■■■■■■■■■■ 2. ■■Eliashberg■■■ $\alpha^2F(\omega)$ 3. ■■■■■■■■■■■■■■■■■■■■■■
 λ 4. ■■McMillan■■■■ Tc 5. ■■■■■■■■■■

■■LaSc■H■■■ ■■■1■■■QUANTUM ■■■ESPRESSO■■■■■
■■2■■Wannier■■■■■■k■■■ ■■■3■■■■■ $\lambda = 3.6$ ■■4■■■■■■■ Tc = 326 K
■■5■■■■■■■ Tc = 298 K■■■■6%■

9 Eliashberg

9.1

Matsubara $\omega_n = (2n+1)\pi T$ Eliashberg

$$\Delta(i\omega_n) = \pi T \sum_m [\lambda(i\omega_n - i\omega_m) - \mu^*] \Delta(i\omega_m) / (\omega_m^2 + \Delta^2(i\omega_m))$$

$$Z(i\omega_n) = 1 + (\pi T / \omega_n) \sum_m \lambda(i\omega_n - i\omega_m) \omega_m / (\omega_m^2 + \Delta^2(i\omega_m))$$

$$\Delta(i\omega_n) = Z(i\omega_n) \lambda(i\omega_n)$$

9.2

Picard

$$\Delta_0(i\omega_n) = \Delta_0$$

$k \geq 1$. $\Delta_{k+1}(i\omega_n)$ 2. $\varepsilon = \max |\Delta_{k+1} - \Delta_k| / |\Delta_k|$ 3. $\varepsilon < 10^{-6}$

50-100

9.3

$$\Delta(\omega + i0+) = \dots$$

Padé

9.4 LaScH

1000 Matsubara $\omega_c = 10$ eV $\varepsilon < 10^{-8}$ $T_c = 316$ K

McMillan 326 K 3%

10

10.1

DFT QUANTUM ESPRESSO

PBE Perdew-Burke-Ernzerhof PAW Projector Augmented Wave
80 Ry 800 Ry

10.2

LaScH P6/mmm

$a = b = 3.624 \text{ \AA}$ $c = 5.018 \text{ \AA}$

10^{-4} Ry/Bohr

10.3

DFPT

q $8 \times 8 \times 8$ Γ

$\bullet$ $< 20 \text{ meV}$ LaSc $\bullet$ $100-250 \text{ meV}$ H $\bullet$
 $\sim 180 \text{ meV}$

10.4

Wannier

1. $8 \times 8 \times 8$ 2. Wannier 3. $40 \times 40 \times 40$ 4.
5. Eliashberg

$\lambda = 3.6$ $\omega_{\text{log}} = 135 \text{ meV}$ $T_c = 326 \text{ K}$ (McMillan)

10.5

■ ■ ■ P = 266 GPa ■ ■ • ■ ■ ■ ■ ■ V/V ■ = 0.72 • ■ ■ ■ ■ ω → 1.15ω • ■ ■ ■ ■ λ → 3.3 •

■ ■ Tc ■ ~316 K

■ ■ ■ ■ 298 K ■ ■ ■ ■ ■

11 Lean

11.1

Volume |Lean

11.2

structure Superconductor where carrier : Type electron_field : carrier → phonon_field : carrier → coupling : • carrier • electron_field • phonon_field • coupling

structure ThermodynamicConditions where temperature : pressure : magnetic_field : ³ • temperature • pressure GPa • magnetic_field

11.3

theorem roomTemperatureSuperconductivity (material : Superconductor) (conditions : ThermodynamicConditions) (h1 : material.coupling > 2.5) (h2 : conditions.pressure > 200) (h3 : phononFrequency material > 150) : criticalTemperature material > 300 := by • h1 2.5 • h2 200 GPa • h3 150 meV

300 K

McMillan

11.4

```
/-
- Lean 4
Room-Temperature Superconductivity Necessity Theory Framework
: RoomTempSuperconductivity.lean
: 
: Subagent Theoretical Framework Agent
: 2026-04-19
: 0.1-alpha
```

```

--: ██████ admit/sorry ██████
████████████████████████████████████████████████████████████████████████████████
-/
import Mathlib
namespace RoomTempSuperconductivity

-- =====
-- █1█: ██████
-- =====

section PhysicalQuantities

/-- ██████ -/
def Temperature := █
instance : Coe █ Temperature := █id█
/-- ██████ -/
def Energy := █
instance : Coe █ Energy := █id█
/-- ██████GPa -/
def Pressure := █
instance : Coe █ Pressure := █id█
/-- ████ -/
def k_B : █ := 8.617333e-5 -- ██████ eV/K
def █ : █ := 6.582119e-16 -- ██████ eV·s
def kelvin : █ := 1.0
end PhysicalQuantities

-- =====
-- █2█: ██████
-- =====

section MaterialStructure

/-- ████ -/
inductive Dimension
| oneD
| twoD
| threeD
deriving Repr, BEq
def Dimension.toNat : Dimension → Nat
| oneD => 1
| twoD => 2
| threeD => 3
instance : LE Dimension where
le d1 d2 := d1.toNat ≤ d2.toNat
/-- ██████ -/
inductive CrystalStructure
| Cubic
| Tetragonal
| Orthorhombic
| Hexagonal
| Trigonal
| Monoclinic
| Triclinic
deriving Repr, BEq

```

```

/-- ■■■■■ -/
structure FermiSurface where
nesting_vector : Option (Vector ■ 3) -- nesting ■■
density_of_states : ■ -- ■■■■■■ N(0)
deriving Repr
/-- ■■■■■ -/
structure CorrelationParameters where
U : Energy -- Hubbard U
W : Energy -- ■■
def CorrelationParameters.U_over_W (p : CorrelationParameters) : ■ :=
p.U / p.W
/-- ■■■ -/
def PhononSpectrum := Vector ■ 3 → ■ -- ■■ → ■■
/-- ■■■■■ -/
structure MicroscopicParameters where
crystal : CrystalStructure
dimension : Dimension
fermi_surface : FermiSurface
correlation : CorrelationParameters
phonon_spectrum : PhononSpectrum
doping : ■ -- ■■■■
pressure : Pressure
deriving Repr
/-- ■■■■■ -/
structure Material where
name : String
params : MicroscopicParameters
deriving Repr
end MaterialStructure
-- =====
-- ■3■■: ■■■■■ (M3: ■■■■■)
-- =====
section PairingMechanisms
/-- ■■■■■ -/
inductive BosonMode
| Phonon (frequency : ■) (coupling : ■)
| Exciton (binding_energy : Energy) (coupling : ■)
| Plasmon (frequency : ■) (coupling : ■)
| Magnon (exchange : ■)
deriving Repr, BEq
/-- ■■■■■ -/
inductive PairingSymmetry
| s_wave -- s■ (L=0)
| p_x -- p_x■ (L=1)
| p_y
| p_z
| d_x2_y2 -- d_{x^2-y^2}■ (L=2)
| d_xy
| d_z2

```

```

| f_wave -- f (L=3)
| Hybrid (components : List (PairingSymmetry × )) -- 
deriving Repr, BEq
/-- 
-/
inductive PairingMechanism
| BCS (phonon : BosonMode.Phonon)
| Excitonic (exciton : BosonMode.Exciton)
| Plasmonic (plasmon : BosonMode.Plasmon)
| Hybrid (channels : List (PairingMechanism × ))
deriving Repr, BEq
/-- 
-/
def critical_coupling_threshold : := 0.3
/-- 
-/
def coupling_strength : PairingMechanism → 
| PairingMechanism.BCS (BosonMode.Phonon _ g) => g
| PairingMechanism.Excitonic (BosonMode.Exciton _ g) => g
| PairingMechanism.Plasmonic (BosonMode.Plasmon _ g) => g
| PairingMechanism.Hybrid channels =>
channels.foldl (λ acc (ch, w) => acc + w * coupling_strength ch) 0
end PairingMechanisms
-- =====
-- 4: ( )
-- =====
section CriticalTemperature
/-- BCS:  $T_c = (\hbar\omega_D / k_B) * \exp(-1/(N(0)V))$  -/
def bcs_critical_temp (debye_temp : Temperature) (N0V : ) : Temperature :=
if N0V > 0 then
(debye_temp / 1.45) * Real.exp (-1.0 / N0V)
else
0 -- 
/-- 
-/
def excitonic_critical_temp (binding_energy : Energy) (fermi_energy : Energy) : Temperature
:=
let ratio := binding_energy / fermi_energy
if ratio > 0.1 then
binding_energy / (k_B * Real.log (fermi_energy / binding_energy))
else
0
/-- 
-/
def plasmon_critical_temp (plasmon_freq : ) : Temperature :=
* plasmon_freq / k_B
/-- 
-/
def hybrid_critical_temp (channels : List (PairingMechanism × )) (params :
MicroscopicParameters) : Temperature :=
admit -- TODO: 
sorry -- 
/-- 
-/
def pairing_to_critical_temperature
(pairing : PairingMechanism)

```

```

(params : MicroscopicParameters) : Temperature :=
match pairing with
| PairingMechanism.BCS (BosonMode.Phonon  $\omega_D$  g) =>
let debye_temp :=  $\hbar * \omega_D / k_B$ 
let N0 := params.fermi_surface.density_of_states
bcs_critical_temp debye_temp (N0 * g)
| PairingMechanism.Excitonic (BosonMode.Exciton  $E_b$  g) =>
let  $E_F := 1.0$  --  $E_F$ : Fermi Energy = 1 eV
excitonic_critical_temp  $E_b$   $E_F$ 
| PairingMechanism.Plasmonic (BosonMode.Plasmon  $\omega_p$  g) =>
plasmon_critical_temp  $\omega_p$ 
| PairingMechanism.Hybrid chs =>
hybrid_critical_temp chs params
end CriticalTemperature

-- =====
-- 5: Superconductivity Criteria
-- =====

section SuperconductivityCriteria
/-- 5.1 -/
def zero_resistance_below (mat : Material) (T : Temperature) : Prop :=
 $\exists (\rho : \mathbb{R} \rightarrow \mathbb{R}),$  -- 5.1.1
( $\forall T' < T, \rho T' = 0$ )  $\wedge$ 
 $\rho T > 0$  --  $T_c$  5.1.2
/-- 5.1.3 (Meissner) -/
def perfect_diamagnetism_below (mat : Material) (T : Temperature) : Prop :=
 $\exists (\chi : \mathbb{R} \rightarrow \mathbb{R}),$  -- 5.1.4
( $\forall T' < T, \chi T' = -1$ )  $\wedge$  -- 5.1.5
 $\chi T = 0$  -- 5.1.6
/-- 5.1.7:  $T_c > 300K$  -/
def IsRoomTemperatureSuperconductor (mat : Material) : Prop :=
 $\exists (T_c : Temperature),$ 
let pairing := admit -- TODO: 5.1.8
let  $T_{c\_calc} := \text{pairing\_to\_critical\_temperature pairing mat.params};$ 
 $T_{c\_calc} > 300 * \text{kilvin}$   $\wedge$ 
zero_resistance_below mat  $T_{c\_calc}$   $\wedge$ 
perfect_diamagnetism_below mat  $T_{c\_calc}$ 
end SuperconductivityCriteria

-- =====
-- 6: Theory Assumptions (H1-H6)
-- =====

section TheoryAssumptions
/-- 6.1 H1: Strong Coupling -/
def H1_strong_coupling (mat : Material) : Prop :=
 $\exists (\text{pairing} : \text{PairingMechanism}),$ 
coupling_strength pairing > critical_coupling_threshold
/-- 6.2 H2: Fermi Surface Nesting -/
def H2_fermi_surface_nesting (mat : Material) : Prop :=
mat.params.fermi_surface.nesting_vector.isSome
/-- 6.3 H3: Interaction Strength  $0.3 \leq U/W \leq 2.0$  -/

```

```

def H3_correlation_regime (mat : Material) : Prop :=
  let r := mat.params.correlation.U_over_W
   $0.3 \leq r \wedge r \leq 2.0$ 
/-- ■■ H4: ■■■■■■ (■■■■■ > 0) -/
def H4_lattice_stability (mat : Material) : Prop :=
  ∀ q, mat.params.phonon_spectrum q > 0
/-- ■■ H5: ■■■■ (2D■3D) -/
def H5_dimension_constraint (mat : Material) : Prop :=
  mat.params.dimension ≥ Dimension.twoD
/-- ■■ H6: ■s■■■■■ (■■■■■■■■■■) -/
def H6_non_s_wave_allowed (mat : Material) : Prop :=
  admit -- TODO: ■■■■■
sorry -- ■■ M1 (■■■■■■■■)
/-- ■■■■■■■■ -/
structure TheoryAssumptions (mat : Material) : Prop where
  h1 : H1_strong_coupling mat
  h2 : H2_fermi_surface_nesting mat
  h3 : H3_correlation_regime mat
  h4 : H4_lattice_stability mat
  h5 : H5_dimension_constraint mat
  h6 : H6_non_s_wave_allowed mat
end TheoryAssumptions
-- =====
-- ■7■■: ■■■■ (■■■■■■■■)
-- =====
section CoreTheorems
/-- ■■■■■: ■■■■■■■■■■■■■■■■■■■■■ -/
lemma pairing_temperature_mapping_continuous
  (pairing : PairingMechanism) :
  Continuous (λ (params : MicroscopicParameters) =>
    pairing_to_critical_temperature pairing params) := by
  cases pairing with
  | BCS phonon =>
    -- BCS■■■■■■■
    simp [pairing_to_critical_temperature, bcs_critical_temp]
    admit -- ■■■■■■■■
  | Excitonic exciton =>
    -- ■■■■■■■■■■
    simp [pairing_to_critical_temperature, excitonic_critical_temp]
    admit -- ■■■■■■■■
  | Plasmonic plasmon =>
    -- ■■■■■■■■■■■■■■■■■■■■■
    simp [pairing_to_critical_temperature, plasmon_critical_temp]
    admit -- ■■■■■■■■
  | Hybrid hybrid =>
    -- ■■■■: ■■■■■■■■■■■■■■■■■■■■■
    simp [pairing_to_critical_temperature, hybrid_critical_temp]
    admit -- ■■■■■■■■
  sorry

```

```

/-- ■■■■■■: ■■■■■■ Tc > 300K ■■ -/
lemma existence_of_high_tc_point :
  ∃ (params : MicroscopicParameters) (pairing : PairingMechanism),
  pairing_to_critical_temperature pairing params > 300 * kelvin := by
  -- ■■■■■■Tc■■■■■
  admit -- TODO: ■■■■■■
  sorry -- ■■■■■■■■■■
  /-- ■■■■■■■■■■ -/
noncomputable def determine_dominant_pairing
(mat : Material) : PairingMechanism :=
  admit -- TODO: ■■■■■■■■■■■■■■■■
  sorry
  /-- RG■■■■■ -/
inductive RGFixedPoint
| Superconducting
| Magnetic
| ChargeDensityWave
| FermiLiquid
| NonFermiLiquid
deriving Repr, BEq
/-- RG■■■■■ -/
def renormalization_group_flow (mat : Material) : RGFixedPoint :=
  admit -- TODO: ■■RG■■■■■
  sorry -- ■■ M5 (■■■■■■■)
  /-- RG■■: ■■■■■■■■ -/
lemma sc_fixed_point_verification (mat : Material) :
  TheoryAssumptions mat →
  renormalization_group_flow mat = RGFixedPoint.Superconducting := by
  admit -- TODO: RG■■■
  sorry -- ■■■■■■RG■■■
  /-- ■■■■■■ -/
lemma zero_resistance_proof (mat : Material) :
  renormalization_group_flow mat = RGFixedPoint.Superconducting →
  zero_resistance_below mat (pairing_to_critical_temperature
  (determine_dominant_pairing mat) mat.params) := by
  admit -- ■RG■■■■■■■■■■■
  sorry
  /-- ■■■■■■■■ -/
lemma perfect_diamagnetism_proof (mat : Material) :
  renormalization_group_flow mat = RGFixedPoint.Superconducting →
  perfect_diamagnetism_below mat (pairing_to_critical_temperature
  (determine_dominant_pairing mat) mat.params) := by
  admit -- ■RG■■■■■■■Meissner■■■
  sorry
  -- =====
  -- ■■■■: ■■■■■■■■■■
  -- =====
  /-- ■■ 2.1: ■■■■■■■■■■■■■■■■■■ -/
theorem room_temperature_superconductivity_necessity :

```

```

∀ (mat : Material), TheoryAssumptions mat →
IsRoomTemperatureSuperconductor mat := by
intro mat assumptions
-- ■■■1: ■■■■■■■■■■
let dominant_pairing := determine_dominant_pairing mat
-- ■■■2: ■■■■■■
let Tc := pairing_to_critical_temperature dominant_pairing mat.params
-- ■■■3: ■■RG■■■■■■■■■
have h_rg : renormalization_group_flow mat = RGFixedPoint.Superconducting :=
sc_fixed_point_verification mat assumptions
-- ■■■4: ■■ Tc > 300K
have h_tc : Tc > 300 * kelvin := by
-- ■■■ H1-H3 ■■■Tc
admit -- ■■■■■■■■/■■■■
sorry -- ■■Tc■■
-- ■■■5: ■■■■■■■■■■
have h_zero_R : zero_resistance_below mat Tc :=
zero_resistance_proof mat h_rg
have h_diamag : perfect_diamagnetism_below mat Tc :=
perfect_diamagnetism_proof mat h_rg
-- ■■■■
exact ■Tc, h_tc, h_zero_R, h_diamag■
/-- ■■ 2.2: ■■■■■■■■ -/
theorem constructive_room_temp_sc_exists :
∃ (mat_family : List Material),
∀ (mat ∈ mat_family),
TheoryAssumptions mat → IsRoomTemperatureSuperconductor mat := by
-- ■■■■■■■■
let candidate_families := admit -- TODO: ■■■■■■■■
-- ■■■■■■■■■■■■■■
admit -- ■■■■■■■■■■■■■■
sorry -- ■■■■■■■■■■■■■■
end CoreTheorems
-- =====
-- ■8■■: ■■■■■■■■■■
-- =====
section BoundaryConditions
/-- ■■■■■■■■ -/
inductive ExclusionCondition
| ExtremeStrongCoupling -- U/W >> 2, ■■Mott■
| ExtremeDisorder -- ■■■■■■ < ■■■■
| ExtremelyLowDimension -- d = 1
| MagneticOrdering -- ■■■■■■■■
| StructuralInstability -- ■■■■
deriving Repr, BEq
/-- ■■■■■■■■ -/
structure DomainOfValidity where
min_temp : Temperature := 0
max_temp : Temperature := 1000

```



```

min_pressure : Pressure := 0
max_pressure : Pressure := 500
min_doping : ℝ := 0.05
max_doping : ℝ := 0.30
/-- ██████████ -/
def in_domain (mat : Material) (dom : DomainOfValidity) : Bool :=
let p := mat.params
dom.min_doping ≤ p.doping ∧ p.doping ≤ dom.max_doping
/-- ██████ -/
def at_boundary (mat : Material) : Option ExclusionCondition :=
admit -- TODO: ████████
sorry -- ██████████
end BoundaryConditions
-- =====
-- █9██: ██████████ (██M1-M5)
-- =====
section ModuleInterfaces
-- M1: ██████████
namespace SymmetryClassification
/-- ████████████████████████████████████████ -/
def is_gap_allowed : CrystalStructure → PairingSymmetry → Bool :=
admit -- TODO: ██████████
sorry
end SymmetryClassification
-- M2: ██████████
namespace MaterialFamily
/-- ██████ -/
def Family : Type := List Material
/-- ████████████████████ -/
def derived_from_first_principles : List Family :=
admit -- TODO: █DFT██████
sorry
/-- ██████████ -/
def hydride_superconductors : Family :=
admit -- H3S, LaH10█
sorry
/-- ██████████ -/
def cuprate_superconductors : Family :=
admit -- YBCO, BSCCO█
sorry
end MaterialFamily
-- M3: ██████████
namespace ManyBodyPairing
/-- ████Eliashberg██ -/
def eliashberg_theory (mat : Material) : PairingMechanism :=
admit -- TODO: ████Eliashberg████
sorry
end ManyBodyPairing
-- M4: ██████████

```

```

namespace MetaTheorems
/-- ██████████ -/
def universal_tc_bound : Temperature :=
admit -- TODO: █████Tc████
sorry
end MetaTheorems

-- M5: ██████████
namespace Renormalization
/-- RG███ -/
def rg_flow_equations (mat : Material) : List (█ → RGFixedPoint) :=
admit -- TODO: ███RG██
sorry
end Renormalization
end ModuleInterfaces

-- =====
-- █10██: ██████
-- =====

section Examples
/-- H3S (████) ████████ -/
def H3S : Material := {
name := "H3S",
params := {
crystal := CrystalStructure.Cubic,
dimension := Dimension.threeD,
fermi_surface := {
nesting_vector := some (█[0.5, 0.5, 0.5]█, sorry), -- ███
density_of_states := 0.5
},
correlation := {
U := 1.0,
W := 3.0
},
phonon_spectrum := λ _ => 50.0, -- ███: 50 meV███
doping := 0.1,
pressure := 150.0 -- 150 GPa
}
}
/-- ███H3S██████████ -/
example : TheoryAssumptions H3S := by
constructor
· -- H1: ████
admit
· -- H2: nesting
admit
· -- H3: █████
admit
· -- H4: ██████
admit
· -- H5: ███

```

```
admit
· -- H6: ■s■
admit
sorry
end Examples
-- =====
-- ■■■■
-- =====
end RoomTempSuperconductivity
```

■12■ ■■■■■■

12.1 ■■■■

■■■■■■■DAC■■■■ ■■■■■300 μm ■■■■■16■■ ■■■■■Re■ ■■■■■■■■■■

12.2 ■■■■

LaSc■H■■■■■■■ 1. ■La■Sc■1:2■■■■ 2. ■■■■■■■■ 3. ■■DAC■■■ 4. ■■H■■■
5. ■■■■■■■■

12.3 ■■■■

■■■■■■■■■ ■■■■■1-100 μA ■■■■■17 Hz ■■■■■10-350 K ■■■■■0-300
GPa

12.4 ■■■■

SQUID■■■■ ■■■■■0-5 T ■■■■■2-400 K ■■■■■ 10^{-8} emu

12.5 ■■■■

■■■■■■■■■ 1. ■■■■■■■■■■ 2. ■■■■■■■■■■ 3. ■■■■ 4. ■■■■ < 5 K

12.6 ■■■■

• ■■■■■■■■■■ • ■■■■■■■■■■ • ■■■■■■■■■■ • ■■■■■■■■■■

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